

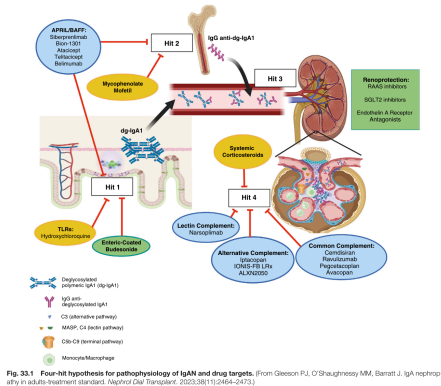


Fig. 33.1 Four-hit hypothesis for pathophysiology of IgAN and drug targets. (From Gleason PJ, O'Shaughnessy MM, Samra J. IgA nephropathy in adults-treatment standard. Nephrol Dial Transplant. 2023;38(11):2464-2473.)

Fig_33_1

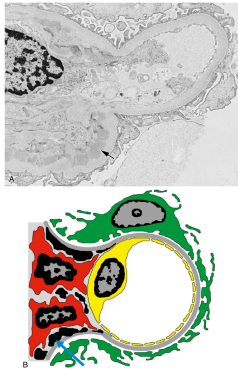


Fig. 33.4 Electron-dense deposits of IgA-containing immune complexes in the paramesangium seen by electron microscopy (A). A cartoon representation of the electron microscopy image is shown in (B).

Fig_33_4

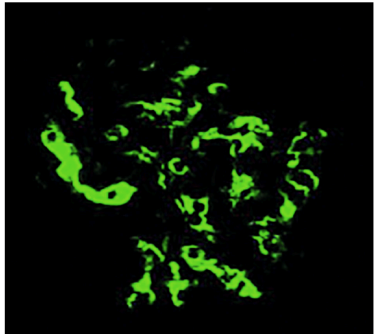


Fig. 33.7 Pericapillary IgA deposition in glomerulus from patient with IgAV.¹⁷⁰

Fig_33_7

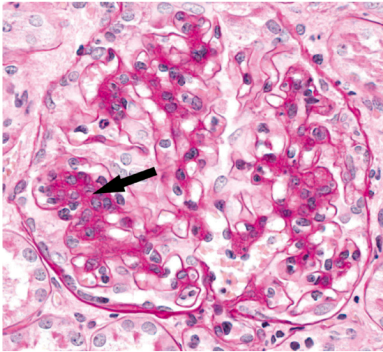


Fig. 33.2 Mesangial hypercellularity in IgAN.

Fig_33_2

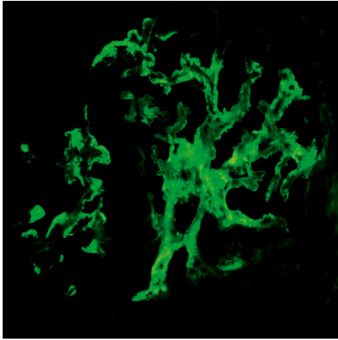


Fig. 33.3 Mesangial IgA deposits detected by immunofluorescent microscopy.

Fig_33_3

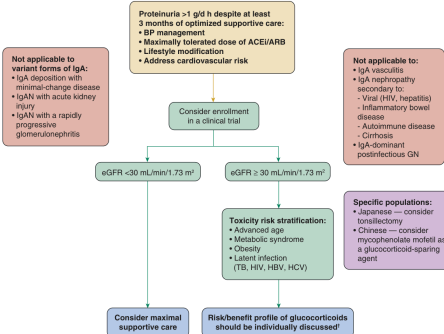


Fig. 33.5 Management of patients with IgAN who remain at high risk for progression after maximal supportive care according to the KDIGO 2021 Clinical Practice Guideline for the Management of Glomerular Diseases. ACEi, Angiotensin-converting enzyme inhibitor; ARB, angiotensin II receptor blocker; BP, blood pressure; eGFR, estimated glomerular filtration rate; GN, glomerulonephritis; HBV, hepatitis B virus; HCV, hepatitis C virus; HIV, human immunodeficiency virus; IgAN, immunoglobulin A nephropathy; TB, tuberculosis. (From KDIGO 2021.¹⁴³)

Fig_33_5

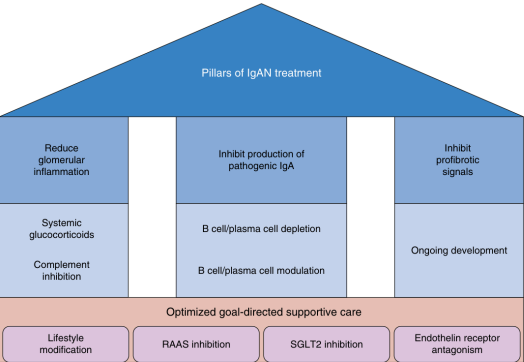


Fig. 33.6 Pillars of IgAN treatment. RAAS, Renin-angiotensin-aldosterone system; SGLT2, sodium-glucose cotransporter-2.

Fig_33_6

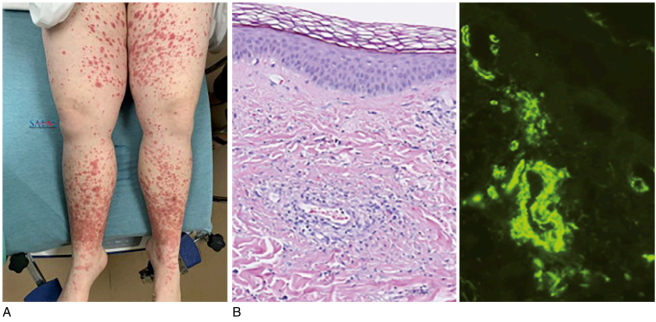


Fig. 33.8 (A) Purpuric rash in IgAV. (B) Leukocytoclastic rash with IgA deposition skin biopsy.¹⁸⁹ (A, Used with the permission of Dr. Erin Shurtlebarger, author, and Megan Schagrin, CEO at SAEM.)

Fig_33_8

Table 33.1 Differential Diagnosis for Glomerular Disease with IgA Deposition	
IgA vasculitis	• Histologically identical to idiopathic IgAN • See Section II for a more detailed discussion of clinical features
Monoclonal gammopathy of renal significance (MGRS)	• Monoclonal IgA-containing deposits less common than monoclonal IgG-containing deposits • Presents histologically as heavy chain deposition disease or proliferative GN with monoclonal immune deposits ¹⁴⁴ • First described in adults with diabetes and Shiga toxin-associated renal insufficiency. The spectrum of disease is heterogeneous and patient phenotypes have been broadened. It is more common >50 years old. ¹⁴⁵ • Typically presents with acute kidney injury, hematuria, and proteinuria in the setting of active infection. • Characteristic histologic features include diffuse proliferative GN with neutrophil infiltration by IgA, dominant C3 staining by IF, and subendothelial and (frequently) subepithelial deposits by IHC. • Mesangial IgA deposition is seen in about 70% of renal medullary cases, ¹⁴⁶ and IgA-dominant cases have been reported. ¹⁴⁷
Lupus nephritis	• Rare cases with subepithelial granular deposition of polymeric IgA have been reported in Asia. ¹⁴⁸ • Disease with a circulating IgA isotype anti-GBM autoantibody and linear deposition of IgA is recognized. ¹⁴⁹ • Given the physiologic role of the renal mesangium in clearance of large molecules from the circulation including polymeric IgA, and the high prevalence of "benign" mesangial IgA deposition, consideration should be given to the possibility of IgA-staining IgA when deposition is found unexpectedly.
Membranous nephropathy	
IgA anti-GBM disease	
Incidental co-deposition of lathyrus IgA	

Table_33_1

Table 33.2 Secondary Causes of IgA Nephropathy	
Clinical Entity	Genetic and Metabolic Features
Lower extremities	- Particularly common in diabetic patients, often as a result of the poor control of glucose level disease.¹⁰ - In adults, histologic IgA deposition is usually accompanied by fibrin or no clinical signs of glomerular disease.
Infantile-onset IgA disease (HSP)	Genetic disease and metabolic defects associated for 75% and 25% of HSP-associated IgA¹¹ respectively in this series.
Celiac disease	ED is improved before IgA in the majority of cases, but in some or active disease, IgA deposition can be identified in up to one third of patients with celiac disease.¹² Although the disease is usually treated in 1-15%, there may also be coexisting dermatitis herpetiformis.¹³
HIV infection	- Glomerular IgA deposits identified in 5% of patients with HIV in one series, and strong association in all cases.¹⁴ - There is a recognized association with IgA.¹⁵⁻¹⁷
Ankylosing spondylitis	IgA is an infrequent extracellular modification.¹⁸ The pathologic changes are similar to those seen in IgA vasculitis.
Rheumatoid arthritis	The risk of IgA increases with severity of joint disease.¹⁹
Psoriasis	- Glomerular IgA deposits are seen in patients with psoriasis concurrently with IgA.²⁰ - Genes reported to present synchronously with diagnosis of psoriasis.
Linear IgA bullous dermatitis	Glomerular IgA deposits are seen in patients with linear IgA bullous dermatitis concurrently with IgA.²¹
Cystic fibrosis	Glomerular IgA deposits are seen in patients with cystic fibrosis.²²
Malnutrition	Glomerular IgA deposits are seen in patients with malnutrition.²³
Drugs	- Anti-TNF-α (cases of IgA have been reported).²⁴ Using the control of IgA TMs, drugs to disease directly associated with IgA (e.g., HSP). - Proteinuria (e.g., HSP) is the most common complication, but associated cases of IgA have also been reported.²⁵

Table_33_2

Table 33.3 MEST-C Score ^{16b,16b}		
Lesion	Score	Description
Mesangial cell proliferation	M0	Average mesangial hypercellularity score ^a <0.5
	M1	Average mesangial hypercellularity score ^a >0.5
Endocapillary hypercellularity	E0	Absent
Segmental	S0	Absent
glomerulosclerosis	S1 ^b	Present
Tubulointerstitial fibrosis	T0	<25%
	T1	26-50%
	T2	>50%
Cellular or fibrocellular crescents	C0	Absent
	C1	Present in at least 1 glomerulus
	C2	Present in >25% glomeruli

^aThe mesangial hypercellularity score is measured as follows: <4 mesangial cells/mesangial area = 0, 4-5 mesangial cells per mesangial area = 1, 6-7 mesangial cells per mesangial area = 2, >8 mesangial cells per mesangial area = 3; the average score of all glomeruli is then calculated.

^bPodocyte hypertrophy/tip lesion characteristics should be reported.

Table_33_3

Table 33.4 International IgAN Prediction Tool Variables^{203a}

Age at time of biopsy
eGFR at time of biopsy
Mean systemic arterial blood pressure at time of biopsy
Proteinuria at time of biopsy (g/day)
Use of RAAS inhibition at time of biopsy
Histologic MEST M-score
Histologic MEST E-score
Histologic MEST S-score
Histologic MEST T-score
Race category: Caucasian, Chinese, Japanese, or Other
Use of immunosuppression preceding biopsy
Interaction effect between proteinuria and T-score
Interaction effect between proteinuria and MAP

Data from Barbour SJ, Coppo R, Zhang H, et al. Evaluating a new international risk-prediction tool in IgA nephropathy. *JAMA Intern Med.* 2019;179:942–952.

Table_33_4

Table 33.5 Treatment Strategies for Patients with IgAN

Clinical Presentation	Treatment	Follow-Up
Isolated nonvisible hematuria ^a	No specific treatment required ^b	Lifelong follow-up and minimum annual BP, eGFR, and UPCR applies to all of the clinical presentations
Episodic visible hematuria ^a	No specific treatment required ^b	
Proteinuria with hematuria	See main text	
Nephrotic syndrome ^c	Treat as minimal-change disease (Chapter 30)	
Rapidly progressive IgAN ^d	Treat with cyclophosphamide and systemic glucocorticoids in accordance with the recommendations for AAV (Chapter 32)	
Malignant hypertension	See Chapter 46	

^aWith normal eGFR.
^bA kidney biopsy is often not performed and a presumptive diagnosis of IgAN is often made. It is important in these cases to consider a type IV collagen disorder.
^cMesangial IgA deposition with light and electron microscopy features otherwise consistent with a podocytopathy resembling minimal-change disease and proteinuria >3.5 g/day with hypoalbuminemia and edema.
^dRapidly progressive IgAN is defined as a 50% decline in eGFR over 3 months, where other causes of rapidly progressive glomerulonephritis (e.g., AAV and anti-GBM disease) and reversible causes (e.g., drug toxicity, common prerenal, and postkidney causes) have been excluded. A kidney biopsy is essential in these cases and will commonly demonstrate mesangial and endocapillary hypercellularity and crescents with areas of focal necrosis. The presence of crescents in a kidney biopsy in the absence of a concomitant change in serum creatinine does not constitute rapidly progressive IgAN; however, these patients require close follow-up to ensure prompt detection of any GFR decline. AAV, ANCA-associated vasculitis; BP, blood pressure; eGFR, estimated glomerular filtration rate; UPCR, urine protein:creatinine ratio.

Table_33_5

Table 33.6 Drugs in Development in 2024 to Target the IgA Nephropathy-Specific Drivers of Nephron Loss

Targeting Pathogenic IgA Synthesis					
Plasma Cell Depletion	Mezagitamab Anti-CD38 Antibody NCT05174221 ^a	Felzartamab Anti-CD38 Antibody NCT05065970			
B cell/plasma cell modulation	Sibeprenlimab APRIL antagonist NCT05248646	Zigakibart APRIL antagonist NCT05852938	Atacicept BAFF/APRIL antagonist NCT04716231	Telitacicept BAFF/APRIL antagonist NCT05799287	Povetacicept BAFF/APRIL antagonist NCT05732402
Inhibiting Glomerular Inflammation					
Alternative complement pathway inhibition	Iptacopan Factor B NCT03373461	IONIS-FB-LRx Factor B NCT05797610	Vemirocan Factor D NCT05097989		
Terminal complement pathway inhibition	Pegcetacopan C3 NCT03453619	ARO-C3 C3 NCT05083364	Ravulizumab C5 NCT04564339	Cemdisiran C5 NCT03841448	Avacopan C5a receptor NCT02384317

^aShows [ClinicalTrials.gov](https://clinicaltrials.gov) identifier.
Data from [http://www.clinicaltrials.gov/ct2/show/study](https://clinicaltrials.gov/ct2/show/study). Drugs in Phase 3 clinical trials in 2024 are in **BOLD**.

Table_33_6

Table 33.7 Sensitivity and Specificity of Different Diagnostic Criteria for IgAV in a Pediatric Population, used to Define the Ankara 2008 Classification^{204a}

Final EULAR/PRINTO/PRES HSP Criteria (with Glossary) and Classification Definition (Sample 973)				
Criterion	Glossary	Sensitivity (%)	Specificity (%)	AUC (%)
Purpura (mandatory criterion)	Purpura (commonly palpable and in crops) or petechiae, with lower limb predominance, ^a not related to thrombocytopenia	89	86	87.5
1. Abdominal pain	Diffuse abdominal colicky pain with acute onset assessed by history and physical examination. May include intrascapular and gastrointestinal bleeding	61	64	62.2
2. Histopathology	Typically leukocytoclastic vasculitis with predominant IgA deposit or proliferative glomerulonephritis with predominant IgA deposit	93	89	91.1
3. Arthritis or arthralgia	Arthritis of acute onset defined as joint swelling or joint pain with limitation on motion	78	42	59.9
4. Renal involvement	Arthralgia of acute onset defined as joint pain without joint swelling or limitation on motion Proteinuria >0.3 g/24 h or >30 mmol/mg of urine albumin/creatinine ratio on a spot morning sample Hematuria or red blood cell casts: >5 red blood cells/high power field or red blood cells casts in the urinary sediment or >2+ on dipstick	33	70	51.4
HSP EULAR/PRINTO/PRES Ankara 2008 classification definition: ≥ 0.90 (95% CI 0.84-0.96)	Purpura or petechiae (mandatory) with lower limb predominance ^a and at least 1 of the 4 following criteria: Abdominal pain, Histopathology, Arthritis or arthralgia, Renal involvement	100	87	93.5

^aFor purpura with atypical distribution, a demonstration of an IgA deposit in a biopsy is required.

AUC, Area under the curve; EULAR, European League Against Rheumatism; HSP, Henoch–Schönlein purpura; PRES, Paediatric

Table_33_7

Table 33.8 ISKDC Histopathologic Classification of IgAV-N^{435a,435b}

ISKDC Grade	Description
Grade I	Minimal alterations
Grade II	Mesangial proliferation
Grade III	Proliferation or sclerosis with <50% crescents ([a] focal or [b] diffuse)
Grade IV	Mesangial proliferation or sclerosis with 50%–75%, crescents ([a] focal or [b] diffuse)
Grade V	Mesangial proliferation or sclerosis with >75% crescents ([a] focal or [b] diffuse)
Grade VI	Membranoproliferative like glomerules

Table_33_8